EXPERIMENT - 5

MODEL DEPLOYMENT AS AN API

Chronic Kidney Disease Classification
Using FastAPI

STUDENT DETAILS

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PUBLIC SOURCE REPOSITORY

https://github.com/kasyap0420/mlops-experiment-5-ckd-prediction-api

Educational model demonstration; not a medical diagnosis.

Docker configuration provided; public API hosting is not claimed.

Experiment 5 — Model Deployment as an API

Chronic Kidney Disease Classification using FastAPI

Evidence generated: 2026-11-01T15:11:40+05:30

Aim

Train a machine learning classification model and serve real-time CKD/Not CKD predictions through a REST API and an interactive dashboard.

Dataset, preprocessing, model

UCI Chronic Kidney Disease (ID 336); 400 records, 24 features, stratified 320/80 training/test split. Numeric missing values: median; scaling: StandardScaler. Categorical missing values: most frequent; encoding: OneHotEncoder. Training preprocessing is fitted inside the scikit-learn pipeline.

Algorithm: **Logistic Regression**; model version: **1.0.0**.

Recorded held-out metric	Value
Accuracy	98.75%
CKD precision	100.00%
CKD recall	98.00%
CKD F1	98.99%
ROC-AUC	99.87%

Metrics come from the existing experiment's saved metadata; they have not been re-estimated on a new dataset.

API endpoints

GET /, GET /health, GET /model-info, POST /predict, GET /metrics, plus optional GET /ready. GET /dashboard serves the additional faculty-required interface. FastAPI Swagger UI is available at /docs.

Testing

The model was loaded from the student's saved Joblib pipeline. The following HTTP checks ran against the local server (actual status codes):

Test	Status	Result
GET service information	200	PASS
GET /health	200	PASS

GET /ready (bonus)	200	PASS
GET /model-info	200	PASS
GET /metrics before tests	200	PASS
GET dashboard	200	PASS
GET dashboard asset app.js	200	PASS
GET dashboard asset styles.css	200	PASS
POST /predict Example A	200	PASS
POST /predict Example B	200	PASS
POST /predict invalid: Missing all required fields	422	PASS
POST /predict invalid: Missing one required field	422	PASS
POST /predict invalid: Invalid feature type	422	PASS
POST /predict invalid: Out-of-range age	422	PASS
POST /predict invalid: Invalid specific gravity	422	PASS
POST /predict invalid: Unexpected extra field	422	PASS
GET /metrics after tests	200	PASS

Sample prediction requests and responses

Example A: demonstration input


```
{
  "age": 50,
  "bp": 80,
  "bgr": 100.0,
  "bu": 36.0,
  "sc": 1.1,
  "sod": 140.0,
  "pot": 4.3,
  "hemo": 14.0,
  "pcv": 42.0,
  "wbcc": 7500.0,
  "rbcc": 4.8,
  "sg": "1.020",
  "al": "0",
  "su": "0",
  "rbc": "normal",
  "pc": "normal",
  "pcc": "notpresent",
  "ba": "notpresent",
  "htn": "no",
  "dm": "no",
```

```
  "cad": "no",
  "appet": "good",
  "pe": "no",
  "ane": "no"
}
```


Example A: actual API response

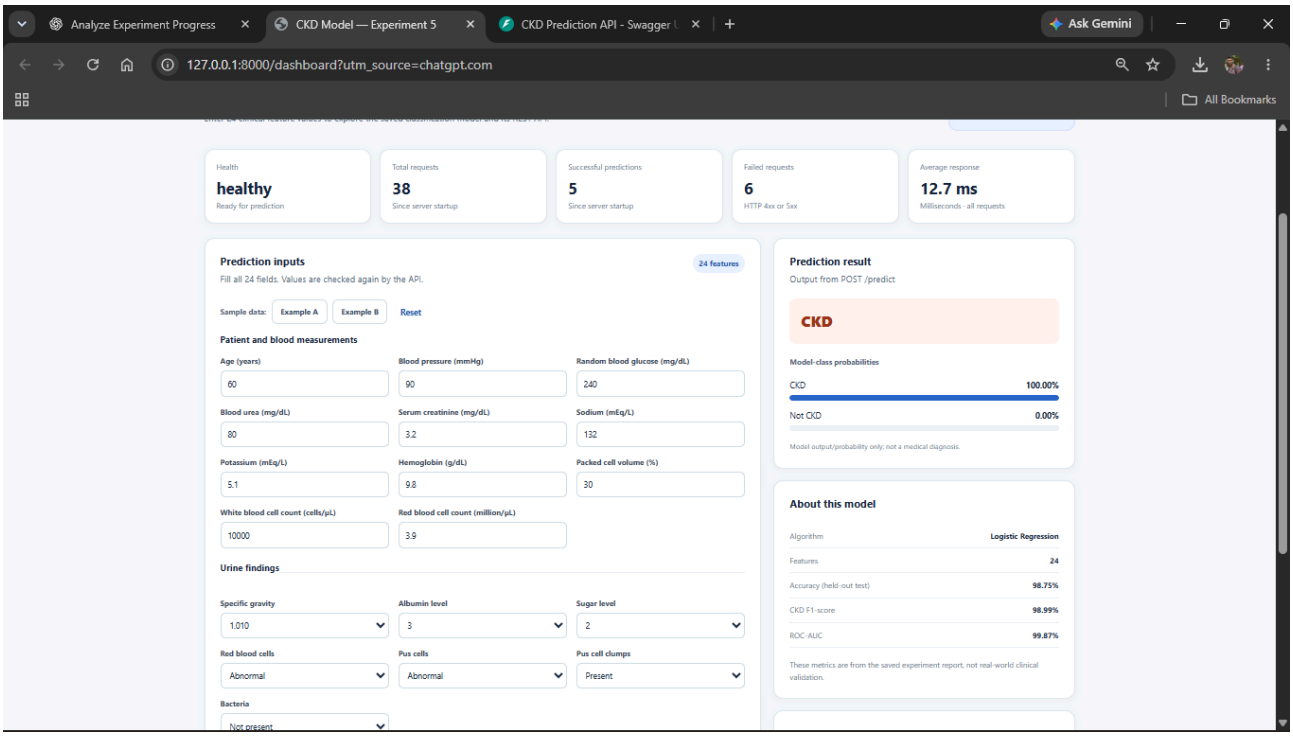

```
{
  "prediction": "Not CKD",
  "model_version": "1.0.0",
  "probabilities": {
    "CKD": 0.18794959575295034,
    "Not CKD": 0.8120504042470497
  },
  "note": "Model output/probability only; not a medical diagnosis."
}
```


Example B: actual API response


```
{
  "prediction": "CKD",
  "model_version": "1.0.0",
  "probabilities": {
    "CKD": 0.9999999945491236,
    "Not CKD": 5.450876385904827e-09
  },
  "note": "Model output/probability only; not a medical diagnosis."
}
```

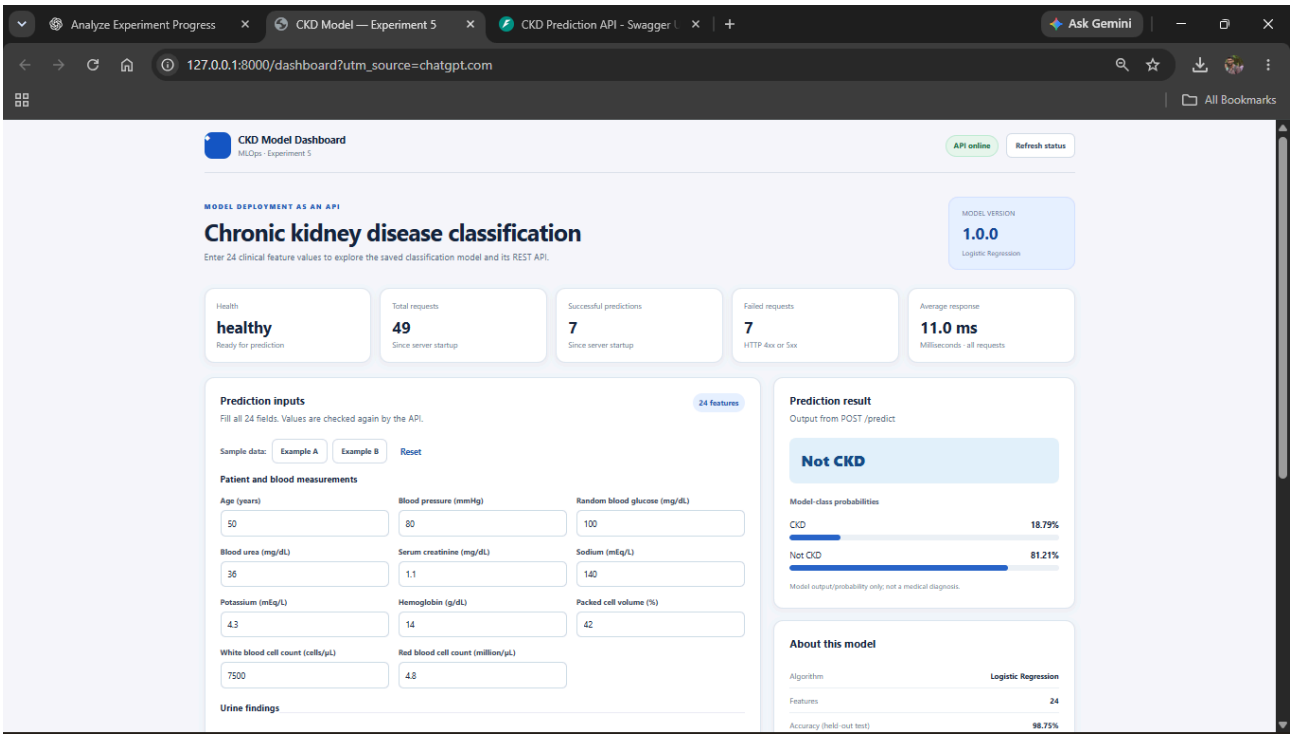

Dashboard evidence

Example B — CKD prediction



Original screenshot supplied by the student during this conversation.

Example A — Not CKD prediction



Original screenshot supplied by the student during this conversation.

Deployment

DEPLOYMENT-READY CONFIGURATION PRESENT; Docker runtime not yet checked

Local service: http://127.0.0.1:8000; dashboard: /dashboard; API documentation: /docs. Deployment-ready Dockerfile and .dockerignore are included. External public deployment is not claimed.

Observations, problems and limitations

The original API/dataset category mismatch was corrected before these tests. Invalid requests return HTTP 422; requests and prediction counts update. Operational metrics are in memory and reset at restart. The dataset and test set are small; predictions and probabilities are educational model outputs, not medical diagnoses.

Full machine-readable requests, responses, invalid test cases and deployment records are in the accompanying evidence folder.

APPENDIX - TEST CASE EVIDENCE

TC-01: CKD prediction 2 page(s)

TC-02: Not CKD prediction 2 page(s)

TC-03: Invalid input (HTTP 422) 1 page(s)

The following pages are the original PDF exports of the test cases.
They contain recorded test results, not independently re-run tests.

Repository:
https://github.com/kasyap0420/mlops-experiment-5-ckd-prediction-api

Experiment 5 — CKD Prediction

CKD Model Deployment as a REST API

Executed: 11/1/2026, 3:32:08 PM

Endpoint: http://127.0.0.1:8000/predict

HTTP Method: POST

PASS — API Test Result

Expected HTTP Status	Actual HTTP Status	Expected Result
200	200	CKD

Actual Prediction

CKD

CKD Probability: **100.00%**

Not CKD Probability: **0.00%**

Submitted Input Features — 24

age **60**	bp **90**	bgr **240**	bu **80**
sc **3.2**	sod **132**	pot **5.1**	hemo **9.8**
pcv **30**	wbcc **10000**	rbcc **3.9**	sg **1.010**
al **3**	su **2**	rbc **abnormal**	pc **abnormal**
pcc **present**	ba **notpresent**	htn **yes**	dm **yes**
cad **no**	appet **poor**	pe **yes**	ane **yes**

Experiment 5 — API Testing Evidence. Synthetic demonstration inputs only. Model probabilities are not medical diagnoses. Results are obtained from the live REST API.

Experiment 5 — Not CKD Prediction

CKD Model Deployment as a REST API

Executed: 11/1/2026, 3:33:13 PM

Endpoint: http://127.0.0.1:8000/predict

HTTP Method: POST

PASS — API Test Result

Expected HTTP Status	Actual HTTP Status	Expected Result
200	200	Not CKD

Actual Prediction

Not CKD

CKD Probability: **18.79%**

Not CKD Probability: **81.21%**

Submitted Input Features — 24

age **50**	bp **80**	bgr **100**	bu **36**
sc **1.1**	sod **140**	pot **4.3**	hemo **14**
pcv **42**	wbcc **7500**	rbcc **4.8**	sg **1.020**
al **0**	su **0**	rbc **normal**	pc **normal**
pcc **notpresent**	ba **notpresent**	htn **no**	dm **no**
cad **no**	appet **good**	pe **no**	ane **no**

Experiment 5 — API Testing Evidence. Synthetic demonstration inputs only. Model probabilities are not medical diagnoses. Results are obtained from the live REST API.

Experiment 5 — Invalid Input Validation

CKD Model Deployment as a REST API

Executed: 11/1/2026, 3:33:42 PM

Endpoint: http://127.0.0.1:8000/predict

HTTP Method: POST

PASS — API Test Result

Expected HTTP Status	Actual HTTP Status	Expected Result
422	422	Validation Error

Validation Error Response

body.age: Input should be greater than or equal to 0

Submitted Input Features — 24

age **-5**	bp **80**	bgr **100**	bu **36**
sc **1.1**	sod **140**	pot **4.3**	hemo **14**
pcv **42**	wbcc **7500**	rbcc **4.8**	sg **1.020**
al **0**	su **0**	rbc **normal**	pc **normal**
pcc **notpresent**	ba **notpresent**	htn **no**	dm **no**
cad **no**	appet **good**	pe **no**	ane **no**

Experiment 5 — API Testing Evidence. Synthetic demonstration inputs only. Model probabilities are not medical diagnoses. Results are obtained from the live REST API.